mojo-adreno · Qwen3-30B-A3B-Instruct-2507

A 30B mixture of experts on a Snapdragon laptop

Int4 Qwen3-30B-A3B as Mojo code on the Adreno X2-90 GPU of a Snapdragon X2 Elite laptop, exact against transformers. Then one of its layers on the laptop’s Hexagon NPU.

31–37
tokens/s decode, 3.3B of 30.5B parameters a token

15.75 GB
on the GPU, inside its 16.8 GB heap

4.31 s
for a 404-token prompt, 2.8× a token at a time

1e-4
logits vs transformers on the same weights

Snapdragon X2 Elite X2E88100 · Windows 11 on Arm · 33.6 GB · Qualcomm Device Cloud, driven from a Mac

Fitting it into the GPU’s half of the RAM

Weights	Format	On the GPU
Experts: 48 × 128 × gate, up, down	int4, a scale per 128	**14.95 GB**
Attention q, k, v, o	int4, a scale per 64	0.48 GB
Output head, 151936 rows	int4, a scale per 64	0.17 GB
Routers and norms	fp32	0.05 GB
KV cache, 512 positions	fp32	0.10 GB
Token embedding	fp32, on the laptop’s disk	none
Total		**15.75 GB**

One 16.82 GB heap

Half the laptop’s RAM, about 16.0 GB of it free. A first layout of 16.7 GB failed to allocate its last 0.95 GB.

Scales searched, not max / 7

Per 128 weights with the least-error scale: 10.7% error on an expert, against 11.2% per 64 with max / 7.

Weights stay on the laptop

vk_server reads them from its own disk: 15.65 GB in 4–6 s (OP_FILE). The tunnel would take 90 minutes.

The laptop downloads the 61 GB bf16 checkpoint and converts it to int4 in 4 minutes · a second heap of 4.29 GB is protected memory

A token, decoded without asking the host

Embedding row
copied from disk by the server (OP_GATHER)

→

Attention
int4 q k v o, QK norm, RoPE, 4 KV heads

→

Router
top 8 of 128 ids and weights, left on the GPU

→

8 experts
int4 gate, up, down, rows found by id

→

Argmax
the next token, into the GPU's token list

Commands a submission	Streamed from the Mac	Replayed on the laptop
1	46.9–49.4 ms a token	**27.2 ms** · 36.7 tokens/s
8	**28.2–32.1 ms** · 31–35 tokens/s	31.4–31.5 ms
64	35.2–38.1 ms	29.6–31.8 ms

531 launches and 1.63 GB of active weights a token: 55–60 GB/s. The server can replay a step on its own clock (OP_RECORD / OP_REPLAY), which times the GPU without the tunnel.

Real weights only: zero-filled ones replayed in 22.5 ms, probably because Windows maps zero pages to shared memory

Prefill when each token picks its own experts

1. **Attention** has the same weights for every token: all the prompt's tokens at once.
2. **Route**: each token's top 8 of the 128 experts.
3. **Group by expert**: each expert gets the tokens that chose it, about 25 of 404.
4. **Grouped GEMM**: each expert's fixed weights are read once, for its tokens only; experts nobody chose cost nothing.
5. **Sum**: each token's 8 outputs, times its router weights.

Prompt	Batched	A token at a time
20 tokens	1.17–1.33 s	**0.77–0.87 s**
48 tokens	**1.26 s**	1.78 s
96 tokens	**1.41 s**	3.61 s
160 tokens	**1.86 s**	5.25 s
404 tokens	**4.31 s**	12.0 s

A lane computes a whole matrix row against 8 tokens, decoding each int4 word once for all 8. Batched from 32 tokens on; shorter prompts leave most of the GPU idle.

Batched logits within 1.5e-5 of a token at a time, the same tokens · 2.6–2.8× from 96 tokens up

The Hexagon NPU: fast hardware, static graphs

One layer, fp16	NPU (HTP V81, ONNX Runtime + QNN)	GPU, int4, all 48 layers
Decode, a token	10.7–14 ms a layer → 0.5–0.7 s a token	**27–32 ms a token**
Prefill, 256 tokens	95–97 ms a layer → 4.6 s	**about 2.8 s**
Prefill, 128 tokens	213–218 ms a layer → 10.3 s	**about 1.6 s**

All 128 experts

A QNN graph is static: it cannot gather a token's 8 experts, so it weights all 128. That is 16× the useful work.

The hardware is fast

fp16 weights at about 100 GB/s and 3.4 TFLOP/s of matrix work. With 8 experts read: about 0.8 ms a layer.

Accurate in fp16

Median token within 1.2e-3 of transformers' layer; QK norms as L2 norms, since QNN's RmsNorm refuses heads of 128.

Too big as fp16

61 GB against 33.6 GB of RAM. The whole model needs the HTP's quantized path (w4a16).

Layer 0 of Qwen3-30B-A3B, real weights, the long prompt's tokens · GPU prefill interpolated from 96, 160 and 404 tokens (1.41, 1.86, 4.31 s)

Checked, and what comes next

Exact on its weights

Logits within 1e-4 of transformers' Qwen3-MoE layers on the same int4 weights; greedy tokens 31 of 31 and 95 of 95.

Close to the bf16 model

31 of 32 greedy choices agree; the one difference comes after the answer ends.

"The sky appears blue because molecules in Earth's atmosphere scatter shorter wavelengths of light, like blue, more effectively than longer wavelengths like red."

The GPU's first sentence, asked why the sky is blue

Next

- Prefill on the matrix units: cooperative-matrix tiles instead of 8-token rows.
- Fuse the norms, router and top 8: a quarter of a step in 192 launches.
- The NPU reading only the chosen experts, or taking attention while the GPU takes the experts.
- Better int4 rounding (GPTQ, AWQ) on the same storage.

github.com/adityaGoCT/mojo-adreno · qwen3moe/ · the reference runs transformers' layers a layer at a time on the laptop